

Linear Equations: One, None, or Infinitely Many Solutions

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

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|-------------|--------------|-------------------|
| 1. 4 | 5. — | 9. , $4x + 8$, 8 |
| 2. — | 6. $17 - 8x$ | 10. — |
| 3. $2x + 6$ | 7. 3 | |
| 4. 7 | 8. $6x - 3$ | |

1. Two saving plans: $3w + 5 = 2w + 9$. Subtract $2w$ from both sides: $w + 5 = 9$. Subtract 5: $w = 4$. The w -terms were different ($3w$ against $2w$), so the variable survived and exactly one value works. Check: $3(4) + 5 = 17$ and $2(4) + 9 = 17$ dollars.

one solution: $w = 4$ weeks

2. Two printers: $4t + 7 = 4t + 2$. The table shows the totals stay 5 pages apart: at $t = 1$ they are 11 and 6, at $t = 2$ they are 15 and 10, at $t = 3$ they are 19 and 14. Algebraically, subtracting $4t$ from both sides leaves $7 = 2$, which is false for every t . Identical t -terms with different constants means the two totals can never meet.

no solution

3. Perimeter two ways: $2(x + 3) = 2x + 6$. Distribute on the left: $2x + 6 = 2x + 6$. Both sides are the same expression, so the statement is true for every value of x — Sam and Dana wrote the same perimeter in two forms.

 $2x + 6 = 2x + 6$, infinitely many solutions

4. $5(x - 2) = 3x + 4$. Distribute: $5x - 10 = 3x + 4$. Subtract $3x$: $2x - 10 = 4$. Add 10: $2x = 14$. Divide by 2: $x = 7$. The x -terms differed, so there is exactly one solution. Check: $5(7 - 2) = 25$ and $3(7) + 4 = 25$.

one solution: $x = 7$

5. $6x - 3(x + 2) = 3x + 1$. Distribute on the left: $6x - 3x - 6 = 3x + 1$. Combine like terms: $3x - 6 = 3x + 1$. Subtract $3x$: $-6 = 1$, which is false. The left and right sides have the same x -term but different constants, so no value of x can work.

no solution

6. $-2(4x - 6) + 5 = 17 - 8x$. Distribute: $-8x + 12 + 5 = 17 - 8x$. Combine the constants on the left: $-8x + 17 = 17 - 8x$. The two sides are the same expression written in a different order, so the equation is true for every x .

 $17 - 8x = 17 - 8x$, infinitely many solutions

7. Two gyms: $25 + 15m = 40 + 10m$. Subtract $10m$: $25 + 5m = 40$. Subtract 25: $5m = 15$. Divide by 5: $m = 3$. The monthly rates differ (15 against 10 dollars), so the costs cross exactly once. Check: $25 + 15(3) = 70$ and $40 + 10(3) = 70$ dollars.

one solution: $m = 3$ months

8. $3(2x - 1) = ax + 7$ with no solution.

(a) Distributing the left side gives

$$6x - 3$$

(b) No solution requires the two sides to have the *same* x -term, so ax must equal $6x$.

$$a = 6$$

(c) **Model explanation (open response — review by hand).** With $a = 6$ the equation reads $6x - 3 = 6x + 7$. Subtracting $6x$ from both sides removes the variable entirely and leaves $-3 = 7$, a statement that is false no matter what x is. Matching x -terms is what erases the variable; different constants are what make the leftover statement false. Both conditions are needed: if the constants had matched as well, every x would work instead of none.

9. Classifying by inspection.

(a) $8x - 3 = 8x + 1$: same x -term, different constants. Subtracting $8x$ leaves $-3 = 1$, false.

$$\text{no solution}$$

(b) $4(x + 2) = 4x + 8$: distributing the left gives $4x + 8$, identical to the right.

$$4x + 8 = 4x + 8, \text{ infinitely many solutions}$$

(c) $5x - 6 = 3x + 10$: different x -terms, so the variable survives. Subtract $3x$: $2x - 6 = 10$; add 6: $2x = 16$; divide by 2: $x = 8$.

$$\text{one solution: } x = 8$$

10. Two ride prices: $2n + 3 = 2n + 5$.

(a) Subtracting $2n$ from both sides leaves $3 = 5$, which is false, so no ride length n makes the prices equal.

$$\text{no solution}$$

(b) **Model explanation (open response — review by hand).** Both companies charge the same amount per mile, so the gap between their prices is the difference in the fixed charges and never changes: Company Q always costs exactly 2 dollars more, whatever the distance. There is no ride length — short, long, or fractional — at which the totals agree, which is exactly what “no solution” says. A rider should simply always choose Company P.